

Case Studies in Data Analytics for Sustainability

Case 4: Recycling - Unit 2

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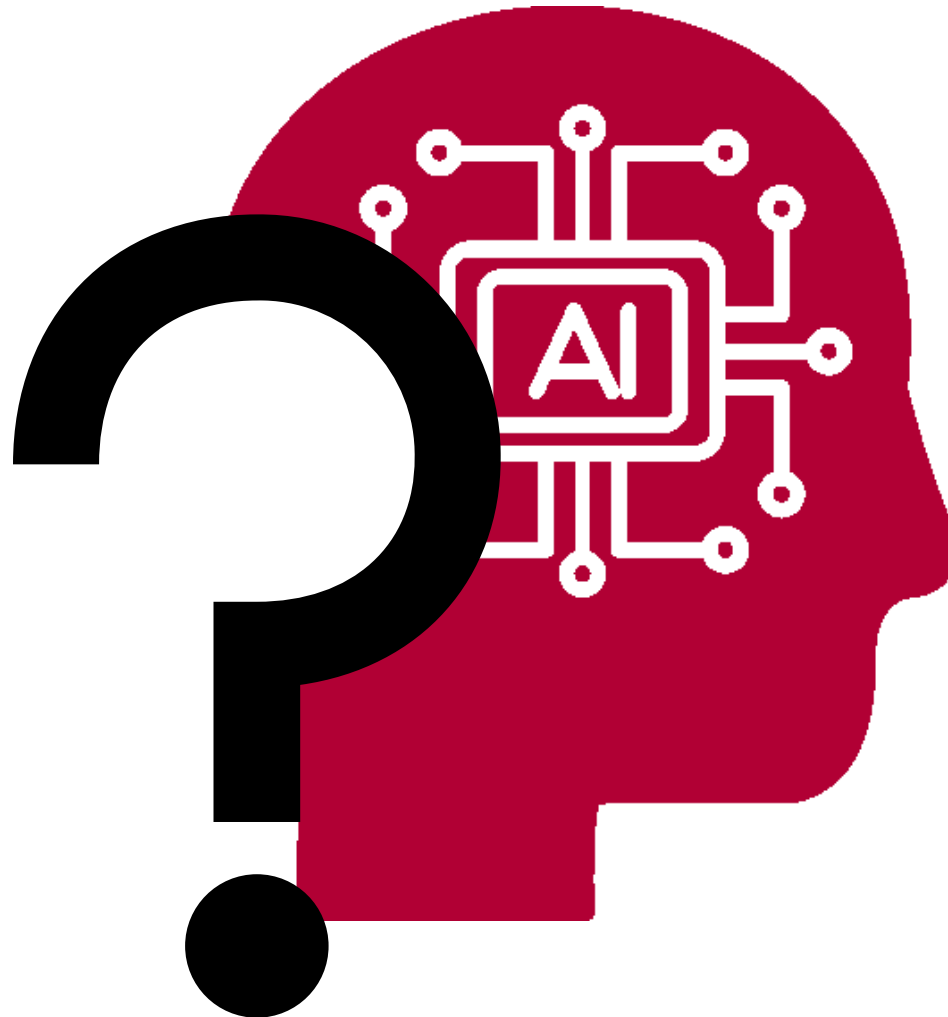


Organizational information



Final Session (Mandatory Attendance)

24.06.2026



How was the last
week?
Any Questions?

Case 4 – Your results



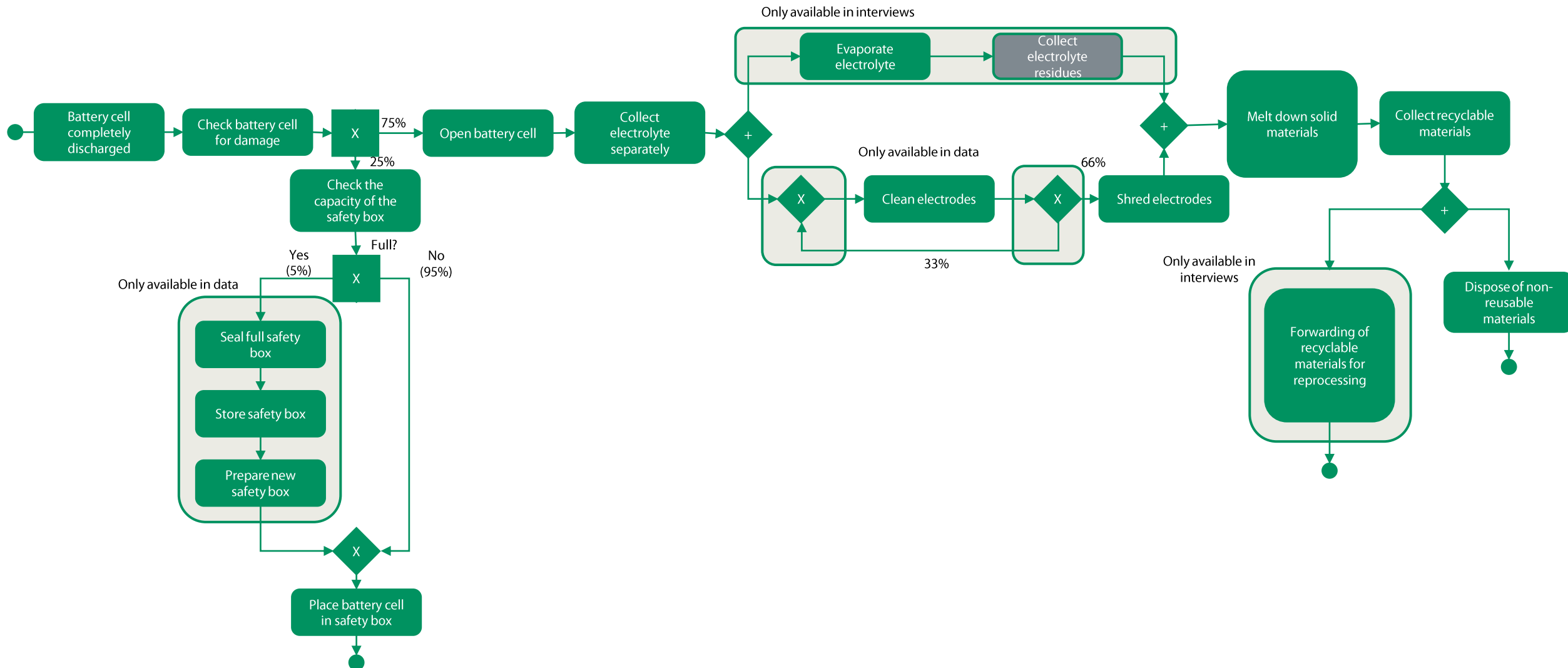
What was done very well

- Looked at the data first
- Good understanding of the BPMN
- Meaningful names for tasks and events



What could have been done better

- Model the interface to the IT department
- Social and Ecological consideration



Case 4: Recycling - Unit 2

Case 1: Material procurement

- What materials should I buy and when?
- Value chain level: Procurement
- Time Series Analysis

Case 2: Predictive Maintenance

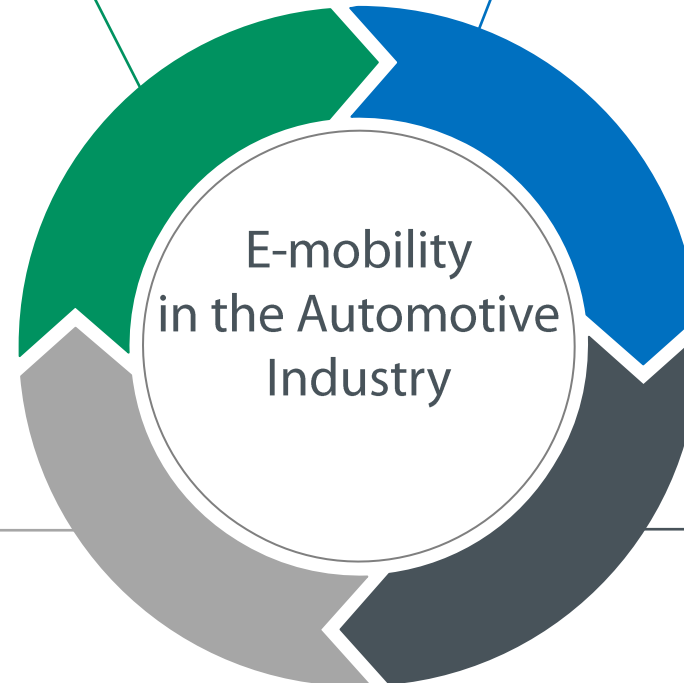
- How often and when should I maintain my machine?
- Value chain level: Operations/production
- Predictive Analytics

Case 4: Recycling

- How much effort do I put into recycling?
- Value chain level: After-sales-services
- Process Mining

Case 3: Quality Management

- How to ensure good quality?
- Value chain level: Operations/production
- Computer Vision





Heat Evaporation

Applying heat to the electrolyte in a controlled environment to vaporize the solvent, leaving behind the lithium salts as solids.



Vacuum Evaporation

Subjecting the electrolyte to a vacuum, which reduces the boiling point of the solvent, allowing it to evaporate at a lower temperature.



Distillation

Heating the electrolyte to its boiling point and collecting the vapor, which is then condensed back into a liquid, leaving behind the salts.



Heat Evaporation

- consumes a significant amount of energy to heat the electrolyte to its boiling point
- cost-effective in terms of equipment and operational costs, yet high energy costs
- easier to train personnel to operate heat evaporation equipment

Vacuum Evaporation

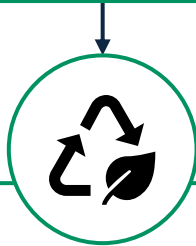
- more energy-efficient than heat evaporation
- equipment can be more expensive to set up and maintain compared to heat evaporation, yet lower energy costs
- may require more specialized training and maintenance expertise

Distillation

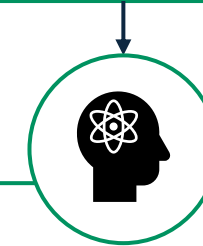
- energy-intensive
- capital-intensive due to the need for specialized distillation equipment
- higher demand for trained professionals in the field, potentially benefiting local communities



The CEO of Edison Cars AG was very satisfied with your process model and that you brought the evaporation of the electrolyte to his attention. He wants you to rethink this process step and figure out the options for the evaporation. The development team identified three methods for evaporation called heat evaporation, vacuum evaporation, and distillation, which all lead to the same result. As you became an expert for battery recycling, management wants you to compare the three methods and recommend one of them for the new facility.



The **CEO aims to know** which **evaporation methods** should be used.



The CEO emphasizes that is very important for him to get a comprehensive **overview on the different sustainability factors**.

1

- Conduct research on each of the three electrolyte evaporation methods
- Understand how the methods work and try to find differences

2

- Gather information on ecological, economic, and social sustainability of each of the evaporation methods
- Use the worksheet provided for the second week to identify relevant numbers

3

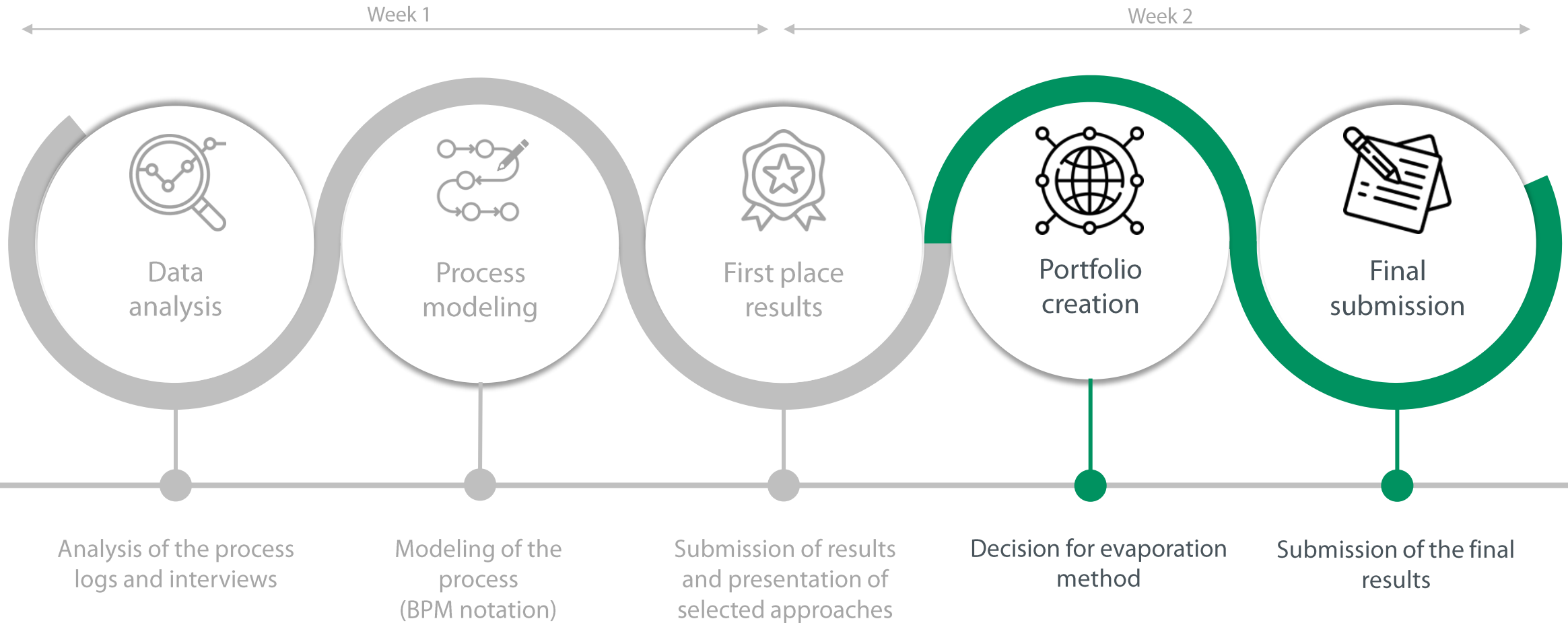
- Decide on the importance of the three types of sustainability for the Edison Cars AG
- Tip: Conduct weighting for each sustainability factor

4

- Compare the sustainability scores you calculated for each of the three methods
- Identify the overall most sustainable method for your battery recycling process and give a recommendation

5

- Prepare a presentation summarizing your findings, including sustainability scores, recommendation and justification
- Prepare to discuss the results with the managers of Edison Cars AG



Worksheet
Contains description of the task
and the exact submission details



**Task
completion**

The following documents must be emailed to **isrenergy@uni-bayreuth.de** as one zip folder by 09:00 AM on 22.06.2026:

Calculation

Description of the steps you did during the evaporation comparison

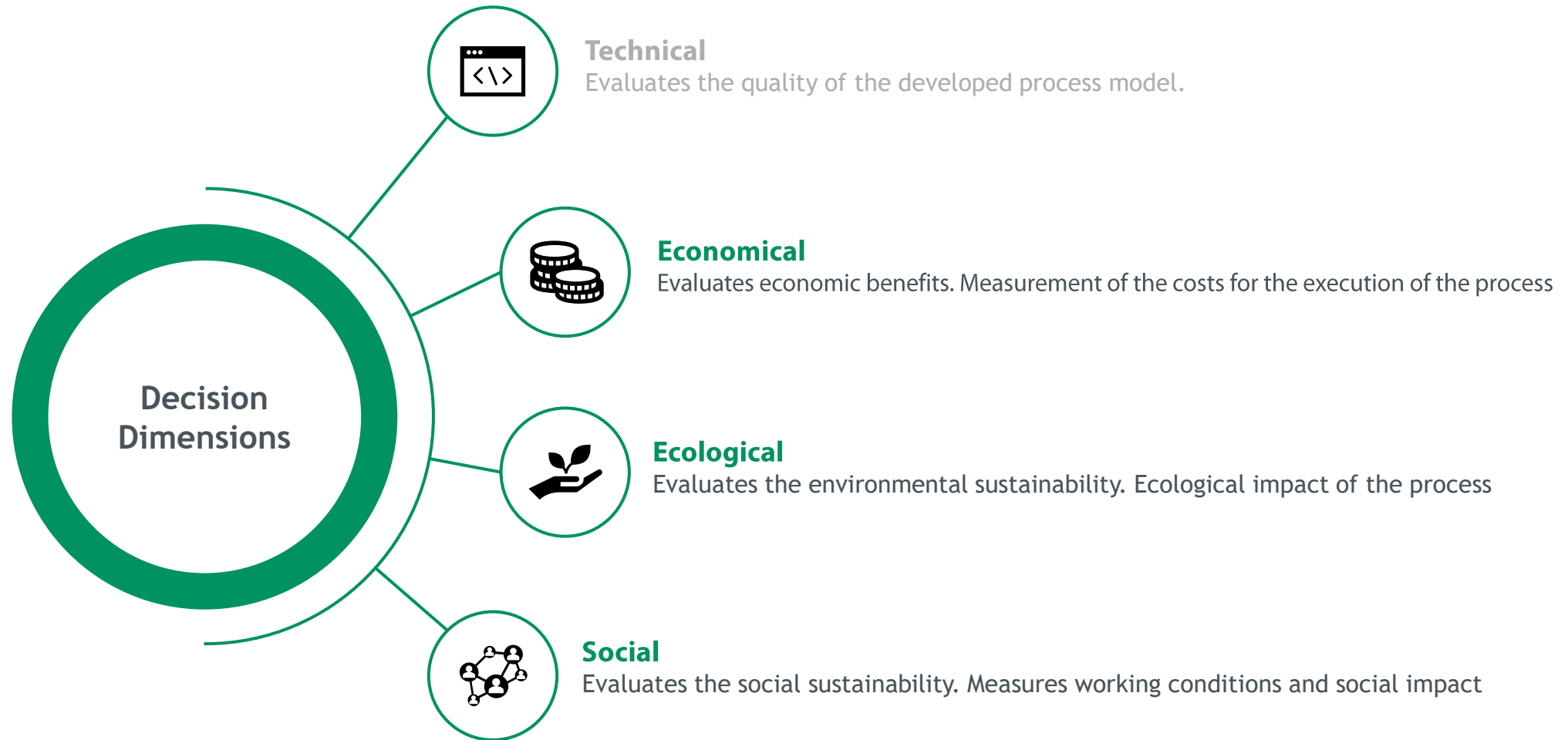


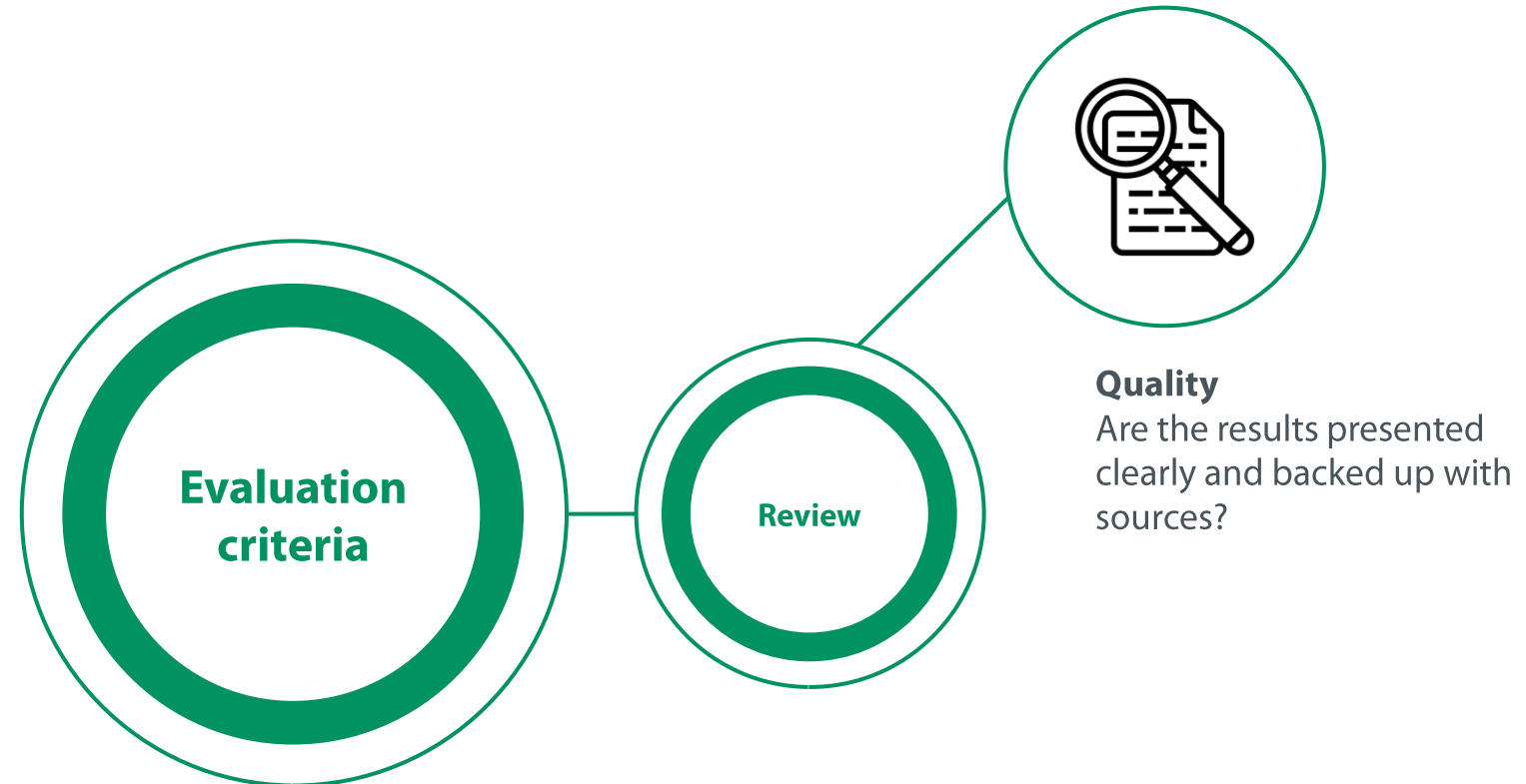
Submission

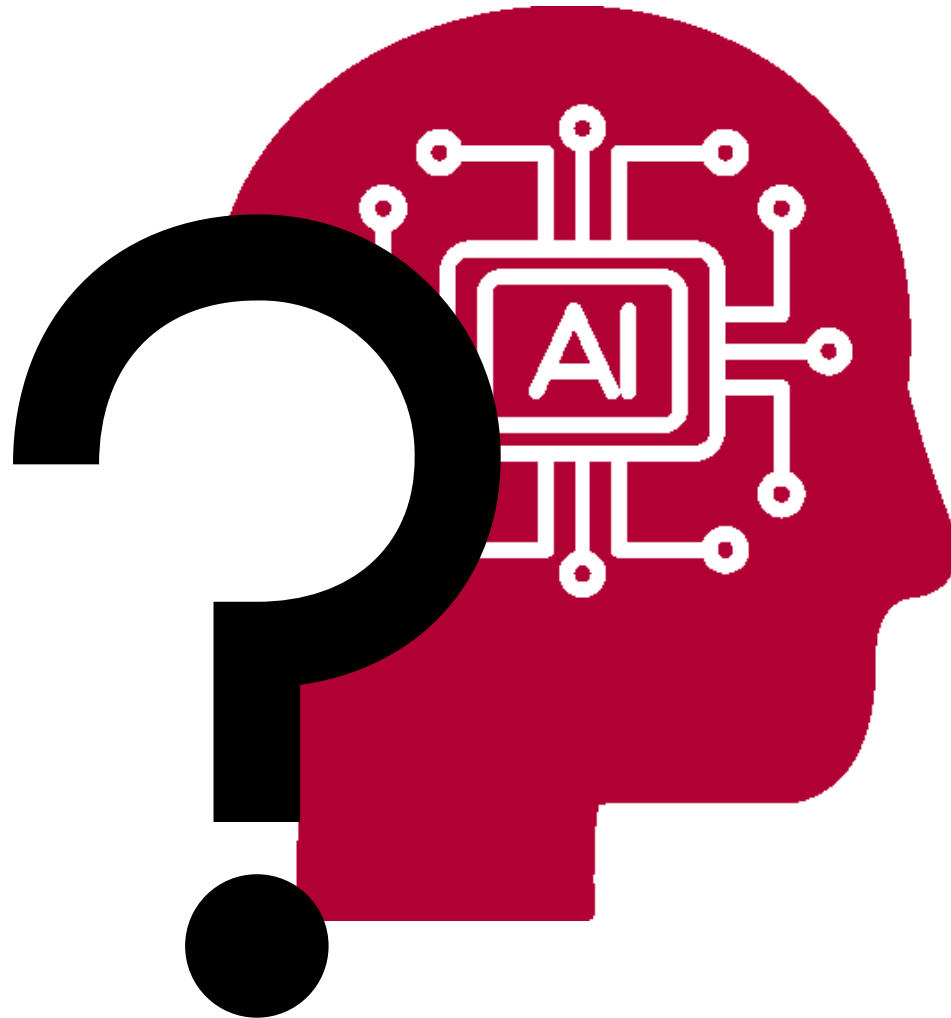


Presentation

A PowerPoint presentation explaining your decisions







Any Questions?